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Disentangled deep learning method for interior tomographic reconstruction of low-dose x-ray CT

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Abstract

Objective. Low-dose interior tomography integrates low-dose CT (LDCT) with region-of-interest (ROI) imaging which finds wide application in radiation dose reduction and high-resolution imaging. However, the combined effects of noise and data truncation pose great challenges for accurate tomographic reconstruction. This study aims to develop a novel reconstruction framework that achieves high-quality ROI reconstruction and efficient extension of recoverable region to provide innovative solutions to address coupled ill-posed problems. *Approach*. We conducted a comprehensive analysis of projection data composition and angular sampling patterns in low-dose interior tomography. Based on this analysis, we proposed two novel deep learning-based reconstruction pipelines: (1) **deep projection extraction-based reconstruction (DPER)** that focuses on ROI reconstruction by disentangling and extracting noise and background projection contributions using a dual-domain deep neural network; and (2) **DPER with progressive extension (DPER-Pro)** that enhances DPER by a progressive 'coarse-to-fine' strategy for missing data compensation, enabling simultaneous ROI reconstruction and extension of recoverable regions. The proposed methods were rigorously evaluated through extensive experiments on simulated torso datasets and real CT scans of a torso phantom. *Main results*. The experimental results demonstrated that DPER effectively handles the coupled ill-posed problem and achieves high-quality ROI reconstructions by accurately extracting noise and background projections. DPER-Pro extends the recoverable region while preserving ROI image quality by leveraging disentangled projection components and angular sampling patterns. Both methods outperform competing approaches in reconstructing reliable structures, enhancing generalization, and mitigating noise and truncation artifacts. *Significance*. This work presents a novel decoupled deep learning framework for low-dose interior tomography that provides a robust and effective solution to the challenges posed by noise and truncated projections. The proposed methods significantly improve ROI reconstruction quality while efficiently recovering structural information in exterior regions, offering a promising pathway for advancing low-dose ROI imaging across a wide range of applications.

1. Introduction

X-ray computed tomography (CT) is one of the most important imaging modalities for clinical diagnosis and plays a key role in modern medicine. Theoretically, analytical and iterative algorithms could provide high-quality CT images when the projection data is sufficient and the reconstruction problem is well-posed (Tuy 1983). However, when taking clinical needs and technical factors into consideration, both imaging

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scanners and raw data are far from being ideal (Wang *et al* 2021). Low-dose interior tomography is a coupled ill-posed problem that integrates low-dose CT (LDCT) with region-of-interest (ROI) imaging. It mainly arises in two scenarios. **(1) Radiation dose reduction.** As radiation exposure is always a concern for CT scans, various strategies have been proposed to lower the radiation dose which mainly fall into two categories: reducing the x-ray flux by adjusting the tube-current-time or tube-voltage (LDCT (Shan *et al* 2019)) and incomplete data acquisition (sparse-view CT (Zhou *et al* 2021), interior tomography (Yu *et al* 2008), and limited-angle CT (Chen *et al* 2022)). In practice, these strategies could be applied jointly to achieve a lower radiation dose. Low-dose interior tomography is a typical example of joint strategies where only a certain ROI of the patient is scanned with a lower x-ray flux. **(2) High-resolution (HR) imaging.** HR tomographic imaging is an indispensable branch for precise diagnosis, for example, the middle and inner ear pathologies (Li *et al* 2020). Equipped with higher-resolution detectors or smaller focal spot sources, higher spatial resolution can be achieved. However, improved resolution is often at the expense of a reduced flux received by each detector element and a smaller field-of-view (FOV). Therefore, many HR scans operate in the interior mode with the FOV focused on ROIs (Li *et al* 2021) and an increased image noise which can also be characterized as low-dose interior tomography.

Low-dose interior tomography is inherently a coupled ill-posed problem as a combination of LDCT and interior tomography which account for highly noisy and truncated projections, respectively. Conventional filtered back-projection (FBP) method suppresses noise in projection using low-pass filtration but often results in smooth structures. Meanwhile, applying FBP to truncated data introduces blooming artifacts, thereby compromising image quality and diagnostic information. Extensive efforts have been made for the reconstruction of LDCT and interior tomography.

For LDCT reconstruction, analytical filtration involves the use of hand-crafted filters for sinogram pre-filtration (Wang *et al* 2005, Manduca *et al* 2009) or image post-processing (Feruglio *et al* 2010, Ma *et al* 2011) to suppress noise. Model-based iterative reconstruction integrates statistical noise models, object properties, imaging physics, and other forms of prior knowledge (Tian *et al* 2011, Zhang *et al* 2015, 2016). Due to the limited capability of filtrations and the large computational cost of the iterative process, these methods remain suboptimal for practical applications (Yang *et al* 2018). Recently, deep learning (DL) methods have yielded remarkable success in addressing various ill-posed problems. Some researchers proposed to incorporate convolutional neural networks (CNN) into projection or image domains (Chen *et al* 2017, Yang *et al* 2018). While these end-to-end pipelines offer computational efficiency, their black-box nature introduces instabilities. Model-based deep learning methods address this by integrating imaging physics and prior knowledge to achieve more robust performance (Wu *et al* 2017, He *et al* 2018). However, these approaches often require large-scale paired normal-dose and LDCT datasets, which are difficult to obtain in clinical settings. This highlights the importance of unsupervised and semi-supervised methods (Lehtinen *et al* 2018, Liang *et al* 2020). Furthermore, recent advances in score-based models including diffusion methods have shown impressive performance in LDCT reconstruction (Gao *et al* 2024, Liao *et al* 2024).

For interior tomography, conventional methods fall into three main categories. The first category involves sinogram extrapolation based on certain assumptions for regions outside the FOV (Hsieh *et al* 2004, Sourbelle *et al* 2005, Van Gompel *et al* 2006). While these methods can mitigate blooming artifacts, the reconstructed attenuation values remain biased due to the inaccuracies in extrapolation. The second category operates under the differentiated back-projection (DBP) framework (Noo *et al* 2004) and an exact solution is proved to be unique with a known subregion (Kudo *et al* 2008). However, such known subregions are not always readily obtainable in practice, limiting their applicability. The third category relies on sparsity models of an ROI using total variation (TV) or high-order TV-regularized iterative reconstruction (Yu and Wang 2009, Katsevich *et al* 2012, Zhang *et al* 2025). Nonetheless, the sparsity prior is often not well satisfied in real-world scenarios which compromises the image quality of reconstructions. For DL-based methods, several studies focus on sinogram extrapolation (Ketola *et al* 2021) while others employ FBP, DBP reconstructions (Han *et al* 2017), or direct back-projected projection (Zhang *et al* 2021) as inputs to CNNs for refinement. Dual-domain methods (Han *et al* 2022) and iterative networks (Ma *et al* 2025) have been explored to further enhance reconstruction quality. Beyond these advancements, the extension of recoverable regions plays a critical role in revealing important anatomical structures outside the ROI (exterior region), which is particularly valuable for diagnosis and localization (Sun *et al* 2025). A plug-and-play method was explored based on data extrapolation from learned prior images (Huang *et al* 2021).

The impressive results of these innovative methods indicate the promise of applying DL techniques to low-dose interior tomographic reconstruction. However, two critical challenges remain to be addressed. First, as revealed by Han *et al* (2022), noise and blooming artifacts exhibit distinct properties that can significantly affect the performance of DL methods. This raises an important question: how can different types of artifacts be effectively disentangled, especially in the context of coupled ill-posed problems? Second,

while sinogram extrapolation is a commonly employed technique in interior tomography, its performance remains suboptimal due to the inherent difficulty in recovering unmeasured information in the exterior regions. Therefore, more effective methods need to be explored for reconstruction from truncated projection, especially when dealing with relatively small ROIs.

In this paper, we propose a disentangled DL framework for low-dose interior tomographic reconstruction. Through a comprehensive analysis of data sampling patterns, we clarified the relationships between different projection components and global data sampling distribution in low-dose interior tomography. Building on these findings, we introduce two novel reconstruction pipelines: (1) **d**eep **p**rojection **e**xtraction-based **r**econstruction (DPER), which focuses on achieving high-quality ROI reconstruction by simultaneously estimating noise and contributions from regions outside the ROI using a dual-domain network architecture; (2) **DPER** with **pro**gressive extension (DPER-Pro), which further enhances reconstruction quality beyond the ROI by performing progressive projection extrapolation. DPER-Pro utilizes projections extracted by DPER and compensates for the missing projections under the guidance of the global angular sampling distribution. Extensive experiments on both simulated and real torso scans demonstrate that the proposed methods outperform existing approaches by delivering superior reconstruction quality within the ROI. Furthermore, DPER-Pro effectively recovers structural information in exterior regions while maintaining high image quality within the ROI.

The contributions of this paper are as follows:

(1) We established a novel DL reconstruction framework for low-dose interior tomography grounded in comprehensive data analysis. Existing LDCT methods cannot directly address the combined challenges of noise and data truncation, and current extrapolation-based routes of corrections often provide limited capability in mitigating non-ideal factors arising inside the FOV. Our framework efficiently mitigates the coupled problem and achieves high-quality ROI reconstruction with efficient extension of recoverable regions.
(2) We proposed an extraction-based pathway DPER that simultaneously estimates projection contributions from exterior regions and noise which offers an efficient and unified solution to the coupled ill-posed problem compared to conventional data extrapolation.
(3) We developed an effective method to extend the recoverable region (DPER-Pro) that disentangles projection components and incorporates progressive data compensation mechanisms. Surpassing existing methods that overlook the underlying data distribution, DPER-Pro improves the image quality of the recoverable region while well preserving the ROI reconstruction quality by leveraging angular sampling patterns to guide a multi-stage projection extension.

The paper is organized as follows: section 2 introduces the imaging model for low-dose interior tomography and provides a comprehensive description of the proposed framework's pipeline. Section 3 presents experimental results on both simulated and real torso data, along with an ablation study to evaluate the effectiveness of the proposed methods. Sections 4 and 5 provide detailed discussions of the results and summarize the key findings and conclusions of the study.

2. Methods

2.1. Problem formulation of interior tomography

Considering the continuous mathematical model of 2D CT imaging, the two-dimensional Radon transform of an object $\mu(x,y)$ (as shown in figure 1(a)) can be expressed as:

$$p_{\mathrm{FULL}}(\theta,t)=\iint \mu(x,y)\,\delta(\mathbf{r}\cdot\boldsymbol{\theta}-t)\,\mathrm{d}x\mathrm{d}y \tag{1}$$

where $\mathbf{r}=(x,y)$ denotes spatial location, $\boldsymbol{\theta}=(\cos\theta,\sin\theta)$ the normal vector of the ray path, and t the distance from the origin to the line of ray path. For interior tomography (see figure 1(b)), only a certain ROI (interior region) $\mu_{\mathrm{ROI}}=\mathcal{I}_R(|\mathbf{r}|)\,\mu$ within an object are targeted to be covered by x-ray beams, e.g. the orange circular region with a radius R and correspondingly the FOV of the imaging system restricted to $t\in[-R,R]$. Hence, the projection from a low-dose interior tomography (interior projection) p_{IN} can be formulated as

$$p_{\mathrm{IN}}(\theta,t)=\mathcal{I}_R(t)\left(p_{\mathrm{FULL}}(\theta,t)+n_{\mathrm{p}}\right) \tag{2}$$

where n_{p} denotes noise and $\mathcal{I}_R(\cdot)$ is an indicator function:

$$\mathcal{I}_R(\alpha)=\begin{cases}1, & |\alpha|\leqslant R\\ 0, & |\alpha|>R\end{cases}. \tag{3}$$

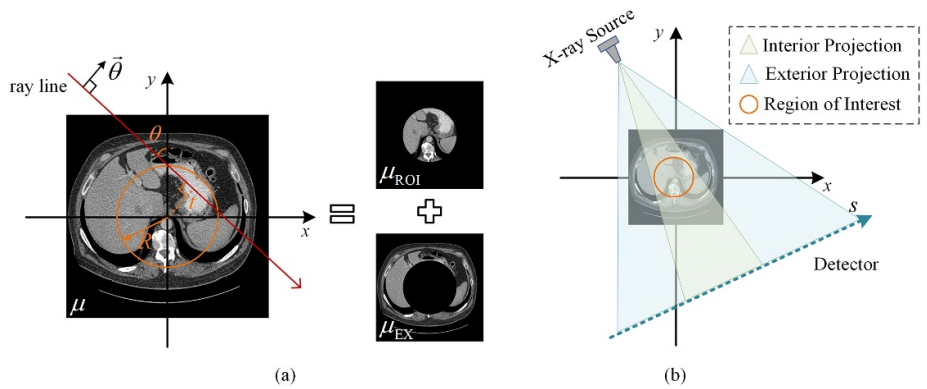


Figure 1. Illustration for interior tomography. (a) Diagram of the interior and exterior regions; (b) diagram of the projection geometry.

Due to data truncation in p_{IN}, directly applying conventional analytical methods results in severe blooming artifacts and intensity deviations in the reconstructions. To address this issue, existing methods often adopt a projection extrapolation strategy to estimate the missing data (exterior projection) p_{EX} outside the FOV $t \in (-\infty, -R) \cup (R, +\infty)$:

$$\begin{aligned} p_{\mathrm{EX}}(\theta, t) &= \iint (1 - \mathcal{I}_R(t))\, \mu(x,y)\, \delta(\mathbf{r}\cdot\boldsymbol{\theta} - t)\, \mathrm{d}x\mathrm{d}y \\ &= \iint \underbrace{(1 - \mathcal{I}_R(t))}_{\text{Outside the FOV}} \underbrace{(1 - \mathcal{I}_R(|\mathbf{r}|))\, \mu(x,y)}_{\text{Exterior Region:}\mu_{\mathrm{EX}}} \delta(\mathbf{r}\cdot\boldsymbol{\theta} - t)\, \mathrm{d}x\mathrm{d}y. \end{aligned} \tag{4}$$

By combining p_{EX} with p_{IN}, a complete non-truncated projection of the object can be achieved. The estimation of p_{EX} can be achieved either through the continuity of projection data (Ohnesorge *et al* 2000, Hsieh *et al* 2004, Van Gompel *et al* 2006, Kolditz *et al* 2011) or DL networks in a data-driven manner (Ketola *et al* 2021, Han *et al* 2022). The objective of projection extrapolation is to estimate a complete projection of the exterior region (outside the ROI) $\mu_{\mathrm{EX}} = (1 - \mathcal{I}_R(|\mathbf{r}|))\mu$. However, when the ROI is small, the information related to μ_{EX} is limited in the interior projection p_{IN}, making the estimation of p_{EX} challenging. Besides, projection extrapolation typically focuses on data compensation outside the FOV while simply neglecting the projections within the FOV. If p_{IN} is contaminated by other factors (such as noise and sparse sampling), additional mechanisms are needed to recover projection data inside the FOV. Therefore, it is crucial to develop reconstruction methods that not only address data truncation but also mitigate other non-ideal factors to achieve high-quality ROI reconstruction.

2.2. Data analysis for interior tomography

Low-dose interior tomography is a coupled ill-posed problem where the projection p_{IN} is influenced by two interconnected aspects:

(1) **Multiple sources of ill-posedness.** p_{IN} is corrupted by both noise and data truncation. If the noise is effectively suppressed, one can achieve a truncated projection as

$$\bar{p}_{\mathrm{IN}}(\theta, t) = \mathcal{I}_R(t)\, p_{\mathrm{FULL}}(\theta, t) \tag{5}$$

(2) **Different projection components from interior and exterior regions** (see figure 2). Given the rays passing through μ_{ROI} only, a complete ROI-only projection p_{ROI} can be generated

$$\begin{aligned} p_{\mathrm{ROI}}(\theta, t) &= \iint \mathcal{I}_R(|\mathbf{r}|)\, \mu(x,y)\, \delta(\mathbf{r}\cdot\boldsymbol{\theta} - t)\, \mathrm{d}x\mathrm{d}y \\ &= \iint \underbrace{\mathcal{I}_R(t)}_{\text{Inside the FOV}} \underbrace{\mathcal{I}_R(|\mathbf{r}|)\, \mu(x,y)}_{\text{Interior Region:}\mu_{\mathrm{ROI}}} \delta(\mathbf{r}\cdot\boldsymbol{\theta} - t)\, \mathrm{d}x\mathrm{d}y. \end{aligned} \tag{6}$$

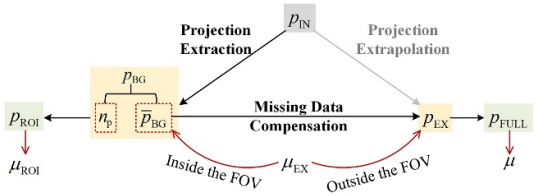


Figure 2. Diagram of the relationships between projection components in interior tomography.

3. Unfortunately, in interior tomography, part of μ_{EX} is also scanned which contributes an additional background projection $\overline{p}_{\mathrm{BG}}$ (see equation (7)) to the projection inside the FOV.

$$\overline{p}_{\mathrm{BG}} = \iint \underbrace{\mathcal{I}_R(t)}_{\text{Inside the FOV}} \underbrace{(1-\mathcal{I}_R(|\mathbf{r}|))\,\mu(x,y)}_{\text{Exterior Region:}\mu_{\mathrm{EX}}} \delta(\mathbf{r}\cdot\boldsymbol{\theta}-t)\,\mathrm{d}x\mathrm{d}y. \tag{7}$$

Thus, $\overline{p}_{\mathrm{IN}}$ can be decomposed into two parts: a complete p_{ROI} from μ_{ROI} and a truncated p_{BG} from μ_{EX}. We have two objectives: (1) achieve high-quality ROI reconstruction; (2) effectively recover structural information of exterior regions. Accomplishing these goals requires efficiently disentangling the coupled ill-posed problem.

For ROI reconstruction, $p_{\mathrm{ROI}}(\theta,t) = \mathcal{I}_R(t)p_{\mathrm{ROI}}(\theta,t)$ is completely within the FOV so that enables high-quality reconstruction. Hence, we propose a projection extraction strategy (see figure 2) which simultaneously separates truncated $\overline{p}_{\mathrm{BG}}$ and noise n_{p} from p_{IN}. The detailed introduction will be provided in section 2.3.

To recover the structural information of the exterior region μ_{EX}, it is essential to further estimate p_{EX} outside the FOV. In fact, $\overline{p}_{\mathrm{BG}}$ and p_{EX} represent the projection components inside and outside the FOV of μ_{EX}, respectively (as shown in figure 2). A feasible approach is to estimate the missing components p_{EX} based on $\overline{p}_{\mathrm{BG}}$. Unlike direct projection extrapolation from p_{IN}, $\overline{p}_{\mathrm{BG}}$ directly captures the projection information of μ_{EX}. At the same time, the interior regions can be well reconstructed through the estimation of $\overline{p}_{\mathrm{BG}}$ and n_{p}. Hence, this pipeline is expected to expand the recoverable regions while ensuring the reconstruction quality of ROIs. Based on this idea, we proposed a progressive decoupled reconstruction method, as detailed in section 2.4.

2.3. DPER

Based on the analysis in section 2.2, p_{IN} can be divided into two parts: the complete ROI-only projection p_{ROI}, and a combination p_{BG} of noise n_{p} and the truncated background projection from exterior regions $\overline{p}_{\mathrm{BG}}$

$$\begin{aligned} p_{\mathrm{BG}}(\theta,t) &= p_{\mathrm{IN}}(\theta,t) - p_{\mathrm{ROI}}(\theta,t) \\ &= \iint \underbrace{\mathcal{I}_R(t)}_{\text{Inside the FOV}} \underbrace{(1-\mathcal{I}_R(|\mathbf{r}|))\,\mu(x,y)}_{\text{Exterior Region:}\mu_{\mathrm{EX}}} \delta(\mathbf{r}\cdot\boldsymbol{\theta}-t)\,\mathrm{d}x\mathrm{d}y + n_{\mathrm{p}}. \end{aligned} \tag{8}$$

Once p_{BG} separated from p_{IN}, the effects of data truncation and noise can be simultaneously eliminated which results in the projection p_{ROI} exclusively covering the ROI. This is the core idea of the projection extraction strategy in this section (as illustrated in figure 3). Unlike existing projection extrapolation strategies, our projection extraction focuses on processing the projection information within the FOV. This approach not only eliminates the contribution of μ_{EX} to the projection inside the FOV to address the data truncation issue, but also suppresses noise and other non-ideal factors. Consequently, it effectively resolves the mixed effects of multiple factors present in interior tomography.

We proposed a novel DPER method for interior tomography. Since our deep learning models work in the discrete data domain, we use $\mathbf{p} \in \mathbb{R}^{N_{\mathrm{D}}N_{\mathrm{A}}\times 1}$ to denote a discretized projection corresponding to p and a matrix $\mathbf{M} \in \mathbb{R}^{N_{\mathrm{D}}N_{\mathrm{A}}\times 1}$ the discretized version of the indicator function $\mathcal{I}_R(t)$ with N_{D} and N_{A} being the numbers of detector bins and view angles, respectively. Hence, the discretized Radon transform of an object $\boldsymbol{\mu} \in \mathbb{R}^{N_{\mathrm{W}}N_{\mathrm{H}}}$ (N_{W} and N_{H} are the width and height of a discretized image) can be expressed as

$$\mathbf{p} = \mathcal{R}(\boldsymbol{\mu}) = \mathbf{H}\boldsymbol{\mu} \tag{9}$$

where $\mathbf{H} \in \mathbb{R}^{N_{\mathrm{D}}N_{\mathrm{A}}\times N_{\mathrm{W}}N_{\mathrm{H}}}$ is a system matrix to formulate the forward projection operator $\mathcal{R}(\cdot)$. Each element $\mathbf{H}_{ij}$ of this matrix represents the contribution of the jth pixel in the imaged object to the ith projection

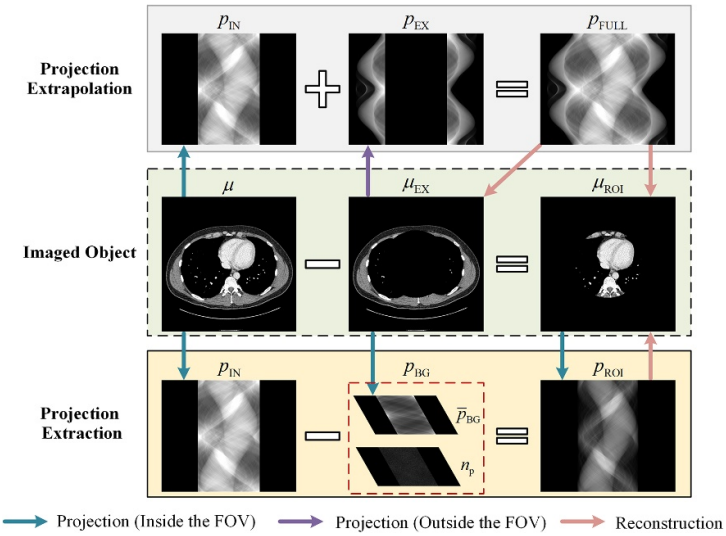


Figure 3. Illustration of two strategies for solving the data truncation problem. The projection extrapolation strategy (first row) estimates the complete projection of the exterior region. Our proposed projection extraction strategy (third row) removes the exterior region's projection contribution within the FOV.

measurement. For image reconstruction, we define an analytical FBP reconstruction operator $\mathcal{R}^{-1}(\cdot)$ in matrix format:

$$\boldsymbol{\mu} = \mathcal{R}^{-1}(\mathbf{p}) = \mathbf{H}_{\mathbf{W}}^{\mathrm{T}}\mathbf{F}\mathbf{W}_{\cos}\mathbf{p} \tag{10}$$

where $\mathbf{W}_{\cos}$ is a diagonal matrix for cosine weighting, $\mathbf{F}$ the Ramachandran Lakshminarayanan (RamLak) filtration, and $\mathbf{H}_{\mathbf{W}}^{\mathrm{T}}$ the weighted back-projection.

DPER is a dual-domain framework (see figure 4(a)): a projection-domain network (PDNet) $\Phi_{\mathrm{P}}(\cdot)$ estimates the noisy background projection $\hat{\mathbf{p}}_{\mathrm{BG}}$ from $\mathbf{p}_{\mathrm{IN}}$. The data are then transformed into the image domain by $\mathcal{R}^{-1}(\cdot)$ to produce an initial reconstruction of the ROI. Finally, an image-domain network (IDNet) $\Phi_{\mathrm{I}}(\cdot)$ further refines details in the reconstructions. As PDNet distinguishes $\hat{\mathbf{p}}_{\mathrm{BG}}$ from $\mathbf{p}_{\mathrm{IN}}$ based on truncation properties, it is specially designed with an expanded receptive field. PDNet and IDNet adopt a U-Net as the backbone where PDNet expands the receptive field of the original U-Net by incorporating three dilated convolution layers with dilation rates of (1, 2, 5) along

$$\hat{\vartheta}_{\Phi_{\mathrm{P}}}, \hat{\vartheta}_{\Phi_{\mathrm{I}}} = \underset{\vartheta_{\Phi_{\mathrm{P}}}, \vartheta_{\Phi_{\mathrm{I}}}}{\arg\min} \left\{ \lambda \frac{\left\| \mathcal{R}^{-1}\left(\Phi_{\mathrm{P}}\left(\mathbf{p}_{\mathrm{IN}}; \vartheta_{\Phi_{\mathrm{P}}}\right)\right) - \boldsymbol{\mu}_{\mathrm{ROI}} \right\|_2}{\left\| \boldsymbol{\mu}_{\mathrm{ROI}} \right\|_2} + \frac{\left\| \Phi_{\mathrm{I}}\left(\mathcal{R}^{-1}\left(\Phi_{\mathrm{P}}\left(\mathbf{p}_{\mathrm{IN}}; \vartheta_{\Phi_{\mathrm{P}}}\right)\right); \vartheta_{\Phi_{\mathrm{I}}}\right) - \boldsymbol{\mu}_{\mathrm{ROI}} \right\|_2}{\left\| \boldsymbol{\mu}_{\mathrm{ROI}} \right\|_2} \right\} \tag{11}$$

the detector axis and (1, 1, 1) along the angle axis in each stage of U-Net (see figure 4(b)). In implementation, the channel number of PDNet and IDNet at each stage was halved compared to the original version ($c = 32$). DPER is trained using dual-domain joint optimization with the loss function defined in equation (11). Here, $\vartheta_{\Phi_{\mathrm{P}}}$ and $\vartheta_{\Phi_{\mathrm{I}}}$ denote the network parameters of PDNet and IDNet, respectively, and λ is a hyperparameter balancing the projection-domain and image-domain supervision.

2.4. DPER with progressive extension of recoverable regions (DPER-Pro)

Besides ROI reconstruction, we also aim to efficiently recover structural information of exterior regions. For a circular ROI with radius R, the angular coverage at each point P: $\mathbf{r} = (x, y)$ in interior tomography is defined as

$$\Theta(\mathbf{r}) = \{\theta \,|\, |\mathbf{r} \cdot \boldsymbol{\theta}| \leqslant R; \theta \in [\theta_{\min}, \theta_{\max}]\} \tag{12}$$

where $[\theta_{\min}, \theta_{\max}]$ denotes the angular range of x-ray sources.

Figure 5 illustrates the global angular coverage distribution achieved by a full-circular interior scan with conjugate projections considered. Within the ROI, angular sampling is complete with $\Theta(\mathbf{r}) = [0, \pi]$. Angular sampling is deficient outside the ROI and drops fast with increasing radial distance from the center. The non-uniform sampling condition complicates data compensation for exterior regions.

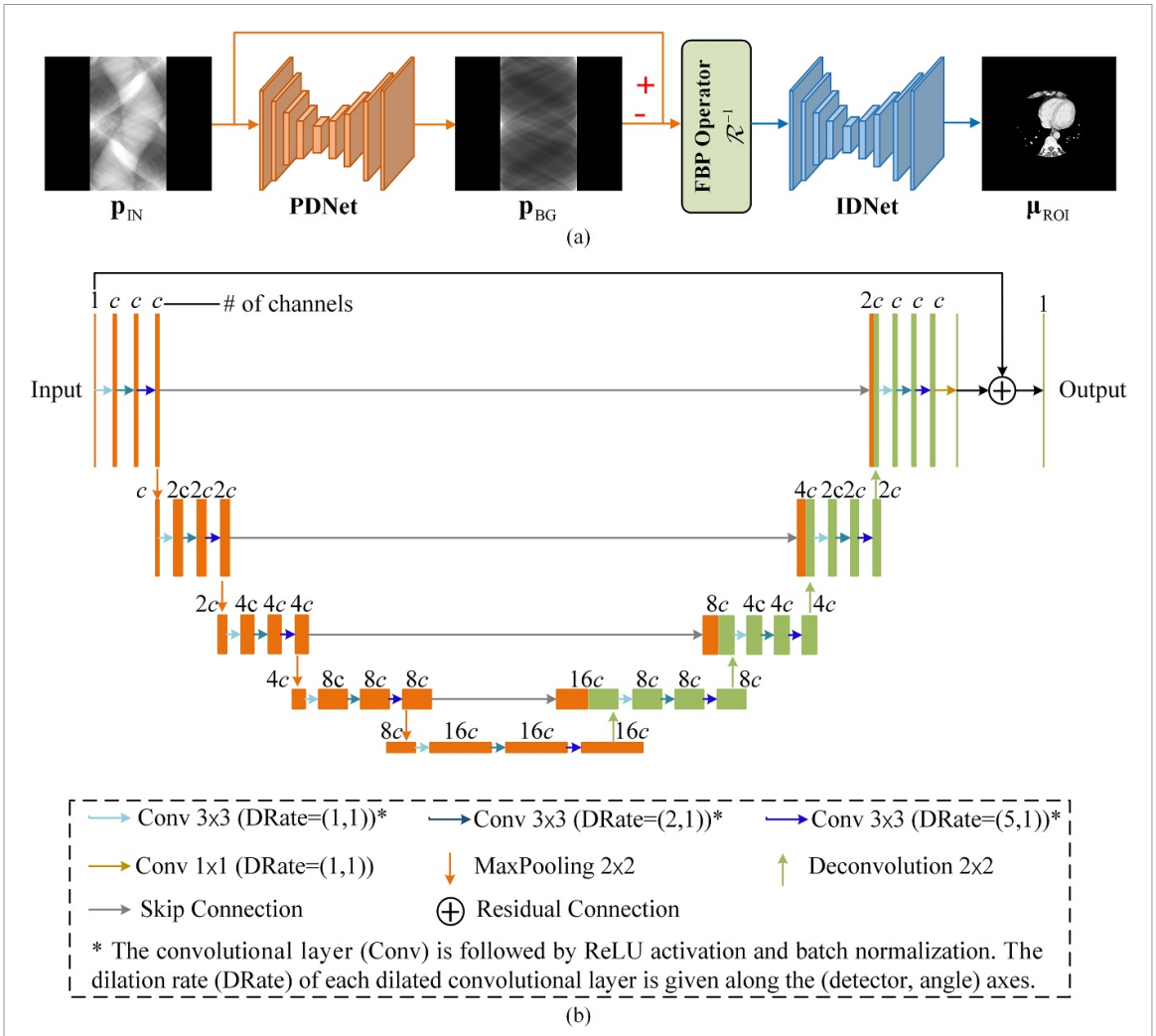


Figure 4. Flowchart of the DPER framework (a) and network architecture of PDNet (b). PDNet estimates the noisy background projection $\hat{\mathbf{p}}_{\text{BG}}$ from the interior projection $\mathbf{p}_{\text{IN}}$. FBP operator generates initial reconstructions that are refined by IDNet.

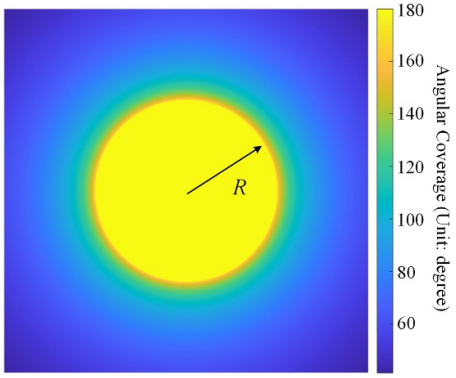


Figure 5. Angular coverage in interior tomography. The angular sampling is complete within the ROI, but angular sampling deficiency increases fast outside the ROI.

To address the coupled ill-posed problem with non-uniform data distribution, we enhance DPER with Progressive extension of recoverable regions (DPER-Pro). As illustrated in figure 6, DPER-Pro includes three projection-domain sub-networks (PDNet1/2/3) and an image-domain network (IDNet). PDNets decouple different projection components in the interior projection step-by-step: PDNet1 removes noise from $\mathbf{p}_{\text{IN}}$ resulting in noise-free truncated projection $\hat{\bar{\mathbf{p}}}_{\text{IN}}$; PDNet2 separates the background projection $\hat{\bar{\mathbf{p}}}_{\text{BG}}$ from $\mathbf{p}_{\text{IN}}$; PDNet3 further estimates the missing projection data $\hat{\mathbf{p}}_{\text{EX}}$ to obtain a complete projection $\hat{\mathbf{p}}_{\text{FULL}} = [\hat{\bar{\mathbf{p}}}_{\text{IN}}; \hat{\mathbf{p}}_{\text{EX}}]$. Here, $[a; b]$ denotes the combined projection data for component a inside the FOV and component b outside

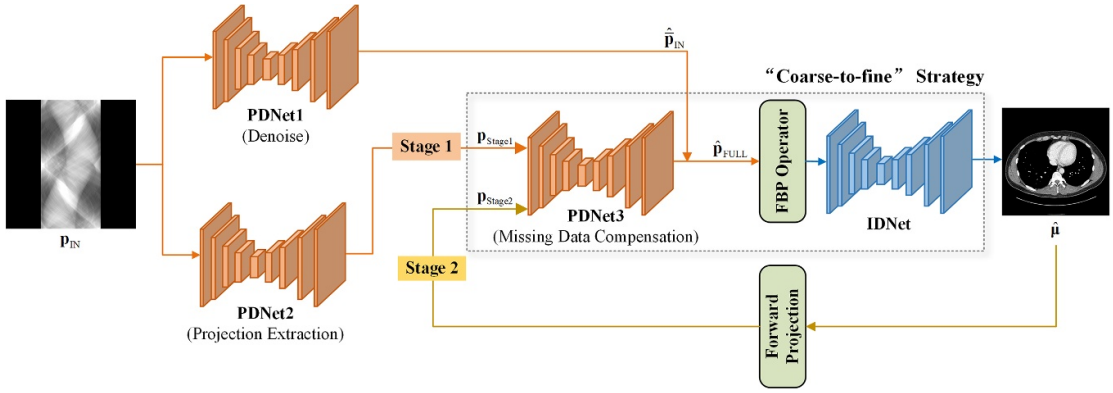


Figure 6. Flowchart of the DPER-Pro framework. PDNets 1/2/3 handle denoising, projection extraction, and missing data compensation, respectively. IDNet enhances the image quality of the final reconstruction. PDNet3 and IDNet utilize a 'coarse-to-fine' strategy: PDNet3 takes the output of PDNet2 as input in the first stage and the forward projection of the first-stage reconstructions as input in the second stage.

the FOV. PDNet3 incorporates positional encoding (Vaswani *et al* 2017) to capture the positional information within completed projections. Then, the analytical operator $\mathcal{R}^{-1}$ generates an initial reconstruction which is refined by IDNet to produce the final reconstruction $\hat{\boldsymbol{\mu}}$. In our implementation, both PDNet2 and PDNet3 utilize a modified U-Net architecture, as illustrated in figure 4(b), with the number of channels set to $c = 16$ and $c = 32$, respectively. PDNet1 and IDNet use the original U-Net as the backbone, with the channel numbers of each stage reduced to one quarter ($c = 16$) and one half ($c = 32$), respectively, to enhance parameter efficiency.

Given the significant variations in data sampling of exterior regions, PDNet3 and IDNet employ a two-stage 'coarse-to-fine' learning strategy. In both stages, PDNet3 integrates $\hat{\bar{\mathbf{p}}}_{\mathrm{BG}}$ inside the FOV (denoted by a projection mask $\mathbf{M}$) with different estimations for regions outside the FOV (denoted by $1-\mathbf{M}$) as inputs $\mathbf{p}_{\mathrm{Stage1}}$ and $\mathbf{p}_{\mathrm{Stage2}}$ (see equation (13)). In the first stage, the initial estimation is provided by a conventional elliptical extrapolation $\hat{\mathbf{p}}_{\mathrm{Ellipse}}$. In the second stage, an improved initial estimation is generated by reprojecting the reconstruction from the first stage $\hat{\boldsymbol{\mu}}_{\mathrm{Stage1}}$ using a forward projection operator $\mathcal{R}(\cdot)$.

$$\begin{cases} \mathbf{p}_{\mathrm{Stage1}} = \left[\hat{\bar{\mathbf{p}}}_{\mathrm{BG}}; \hat{\mathbf{p}}_{\mathrm{Ellipse}}\right] \\ \mathbf{p}_{\mathrm{Stage2}} = \left[\hat{\bar{\mathbf{p}}}_{\mathrm{BG}}; (1-\mathbf{M}) \odot \mathcal{R}\left(\hat{\boldsymbol{\mu}}_{\mathrm{Stage1}}\right)\right] \end{cases}. \tag{13}$$

The progressive projection restoration is guided by angular coverage $\Theta(\mathbf{r})$. Specifically, a threshold θ_n is defined to identify a priority region $\Upsilon_n(\Theta(\mathbf{r}), \theta_n)$ for the nth stage

$$\Upsilon_n(\Theta(\mathbf{r}), \theta_n) = \{\mathbf{r} | \Theta(\mathbf{r}) \geqslant \theta_n\}. \tag{14}$$

Then, a weighting function $g_n(\mathbf{r}, \theta_n)$ is applied to the image-domain loss with a piecewise linear function combined with exponential decay (see equation (15)). This assigns higher weights to the prioritized region than the non-prioritized regions:

$$g_n(\mathbf{r}, \theta_n) = \begin{cases} 1, & (\Theta(\mathbf{r}) > k_1\theta_n) \\ \mathrm{e}^{\frac{-2(\Theta(\mathbf{r})-k_1\theta_n)^2}{(k_1-k_2)^2\theta_n^2}}, & (\Theta(\mathbf{r}) \in [k_2\theta_n, k_1\theta_n]) \\ \mathrm{e}^{-2}, & (\Theta(\mathbf{r}) < k_2\theta_n) \end{cases} \tag{15}$$

where k_1 and k_2 are the parameters to control the transition region. In this way, DPER-Pro focuses on the priority regions with the angular coverage exceeding θ_1 and θ_2 ($\theta_2 < \theta_1$) during the two-stage data recovery. After the first stage, a forward projection operator $\mathcal{R}$ maps the image back into the projection domain which provides a reliable initial estimate for the second stage. In the training process, the two-stage DL models share network parameters to reduce complexity. The overall loss function (see equation (16)) consists of four parts: denoising loss L_{PDNet1}, projection extraction loss L_{PDNet2}, missing data compensation loss L_{PDNet3}, and image recovery loss L_{IDNet}. Three hyperparameters $\lambda_{\mathrm{PDNet1}}$, $\lambda_{\mathrm{PDNet2}}$, and $\lambda_{\mathrm{PDNet3}}$ balance these terms which ensure their magnitudes (L_{PDNet1}, L_{PDNet2}, and L_{PDNet3}) are approximately equal. Their combined magnitude

is close to that of the image recovery loss L_{IDNet} for balanced emphasis across all subtasks.

$$\text{Loss} = L_{\text{IDNet}} + \lambda_{\text{PDNet1}} L_{\text{PDNet1}} + \lambda_{\text{PDNet2}} L_{\text{PDNet2}} + \lambda_{\text{PDNet3}} L_{\text{PDNet3}}$$

$$\begin{cases} L_{\text{IDNet}} = \sum_{n=1}^{N_{\text{Stage}}} \frac{\|(\hat{\boldsymbol{\mu}}-\boldsymbol{\mu})g_n(\mathbf{r},\theta_n)\|_2}{\|\boldsymbol{\mu}g_n(\mathbf{r},\theta_n)\|_2} \\ L_{\text{PDNet1}} = \frac{\left\|\hat{\bar{\mathbf{p}}}_{\text{IN}}-\bar{\mathbf{p}}_{\text{IN}}\right\|_2}{\|\bar{\mathbf{p}}_{\text{IN}}\|_2} \\ L_{\text{PDNet2}} = \frac{\left\|\hat{\bar{\mathbf{p}}}_{\text{BG}}-\bar{\mathbf{p}}_{\text{BG}}\right\|_2}{\left\|\bar{\mathbf{p}}_{\text{BG}}\right\|_2} \\ L_{\text{PDNet3}} = \sum_{n=1}^{N_{\text{Stage}}} \frac{\left\|\left(\mathcal{R}^{-1}\left(\hat{\mathbf{p}}_{\text{FULL}}\right)-\boldsymbol{\mu}\right)g_n(\mathbf{r},\theta_n)\right\|_2}{\|\boldsymbol{\mu}g_n(\mathbf{r},\theta_n)\|_2} \end{cases} . \quad (16)$$

3. Experimental results

3.1. Experimental set-up

The proposed methods were evaluated on both a simulated torso dataset and a practical CT scan of a torso phantom under a circular fan-beam geometry. The analytical reconstruction operator $\mathcal{R}^{-1}(\cdot)$ and forward projection operator $\mathcal{R}(\cdot)$ were implemented as a fixed linear layer in the network using a home-made toolkit. In the training process, we set $\lambda = 1$ for the DPER loss function (equation (11)). DPER-Pro adopts a two-stage learning mechanism with angular thresholds θ_1 and θ_2 being $2\pi/3$ and $\pi/2$ in equation (14). Weighting function parameters k_1 and k_2 (equation (15)) are set to 1.2 and 0.8, respectively. For the DPER-Pro loss function (equation (16)), we set λ_{PDNet1}, λ_{PDNet2}, and λ_{PDNet3} to 1, 1, and 0.1, respectively.

To validate performance, we compared the proposed methods against four representative approaches: (1) FBP reconstruction; (2) FBPConvNet based on an image-domain UNet; (3) DD_ExDenoise (Han *et al* 2022): a representative and dedicated algorithm specifically designed to address both noise and truncation issues in low-dose interior tomography. It employs a dual-domain DL method with dual-branch output for projection denoising and extrapolation. (4) DL_PnP (Huang *et al* 2021): an iterative reconstruction method using DL network predictions as priors under a plug-and-play framework to extend the FOV. Quantitative evaluation was conducted by calculating the mean and coefficient of variation (CV) for the structural similarity index (SSIM) and peak signal-to-noise ratio (PSNR) across all test slices.

3.2. Simulation experiments

Detailed geometric parameters for the simulated low-dose interior tomography are listed in table 1. The simulation used open-source torso datasets from the American Association of Physicists in Medicine (AAPM) 'low dose grand challenge' (McCollough 2016) and the Cancer Imaging Archive (TCIA) 'Low Dose CT Image and projection data' (Moen *et al* 2021). To address the difference in slice thickness between the two datasets, we selected every fifth slice from the AAPM dataset to reduce the effect of nonuniform thickness across both datasets. A total of 4066 slices from 29 patients were used for training, 758 slices from 5 patients for validation, and 757 slices from 5 patients for testing. Poisson noise was simulated with 10^5 incident photons per ray.

3.2.1. ROI reconstruction

ROI reconstruction results for three test samples under the simulated projection noise level of the training data (Poisson noise with 1×10^5 incident photons) are presented in figure 7. FBP reconstructions suffered from severe intensity deviation (see figure 7(b)) which were corrected by FBPConvNet. However, the FBPConvNet results remained blurry (see figure 7(c)). DDEx_Denoise effectively restored intensity and suppressed noise but left artifacts and hallucinations in the reconstructions (see figure 7(d)), indicating its limitations in addressing the challenges of coupled ill-posed problems. In comparison, DPER and DPER-Pro achieved the best reconstruction quality with clear visibility of detailed structures (see figures 7(e) and (f)). The quantitative metrics for the whole test set (summarized in table 2) confirmed the advantage of DPER and DPER-Pro over the comparative methods. We also investigated the number of parameters and training time required to achieve the reported best-validation models. The cross-domain methods (DD_ExDenoise, DPER, and DPER-Pro) utilize fewer parameters than FBPConvNet while achieving superior results, demonstrating higher parameter efficiency. However, the incorporation of domain transformation operators $\mathcal{R}^{-1}$ and $\mathcal{R}$ in the cross-domain methods can lead to increased training time as shown in table 2.

The methods were further evaluated under higher noise (Poisson noise with 5×10^4 incident photons). All DL models were finetuned using data from one patient in the training set at this noise level. Figure 8 presents the results of the second sample from figure 7. The quantitative metrics for the whole test set are

Table 1. Geometric parameters for the simulated interior tomography.

Parameters	Value
Distance from x-ray Source to the Isocenter	410 mm
Distance from detector to the Isocenter	310 mm
Number of detector elements—ROI	512
Number of detector elements—Full	1056
Size of detector elements	0.45 mm
Dimension of reconstruction grids—ROI	256 × 256
Dimension of reconstruction grids—Full	512 × 512
Pixel size	0.5 mm
Number of projection views over 2π	480

Table 2. Quantitative metrics for all test slices (Poisson noise with 1×10^5 incident photons).

			SSIM		PSNR	
Metric	Time (h)	Parameter	Mean	CV (%)	Mean	CV (%)
FBP	—	—	0.4898	20.10	12.38	30.01
FBPConvNet	13.7	35 M	0.9431	3.488	37.43	**6.483**
DD_ExDenoise	21.6	17 M	0.9605	2.804	39.77	8.764
DPER	19.7	21 M	0.9680	**2.509**	40.98	6.998
DPER-Pro	55.7	27 M	**0.9681**	2.514	**41.77**	7.151

Note: The best results are highlighted in bold.

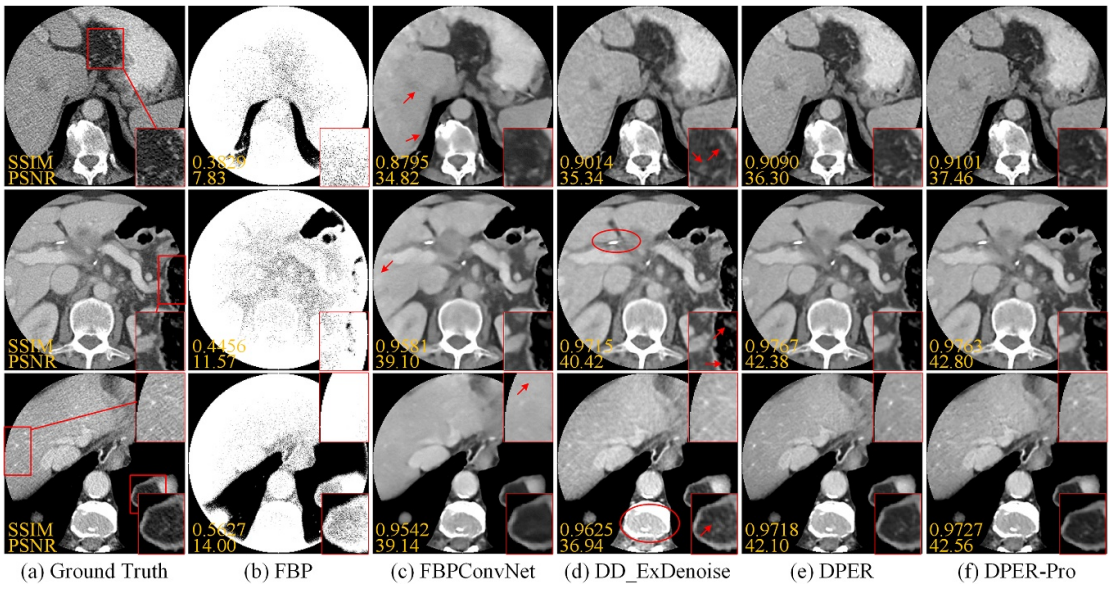


Figure 7. ROI reconstructions from simulated experiments (Poisson noise with 1×10^5 incident photons): ground truth (a), FBP (b), FBPConvNet (c), DD_ExDenoise (d), DPER (e), and DPER-Pro (f). The display window is [0.016, 0.024]. SSIM and PSNR are in the bottom-left corner of each image.

summarized in table 3. FBPConvNet struggled to adapt to higher noise, resulting in significant intensity deviation (see figure 8(a)). While DD_ExDenoise effectively suppressed noise, residual artifacts and noticeable deviations in intensity remained in the reconstructions (see figure 8(b)). In contrast, both DPER and DPER-Pro achieved superior reconstruction quality qualitatively and quantitatively. However, DPER showed minor artifacts (see the red ellipse in figure 8(c)), whereas DPER-Pro suppressed noise and restored detailed structure better. These results confirmed that DPER and DPER-Pro adapted well to higher noise levels with DPER-Pro achieving the best performance.

3.2.2. Extension of recoverable regions

According to the geometric parameters in table 1, the ROI diameter is 128 mm (256 pixels). In the exterior regions, a 150 mm diameter circular region (300 pixels, denoted as 'FOV300') with a minimum angular coverage of 120 degrees can be effectively reconstructed based on prior limited-angle CT experience (Chen

Table 3. Quantitative metrics for all test slices (Poisson noise with 5×10^4 incident photons).

	SSIM		PSNR	
Metric	Mean	CV	Mean	CV
FBPConvNet	0.9182	3.888%	32.51	6.510%
DD_ExDenoise	0.9533	2.894%	37.53	**6.128%**
DPER	0.9587	**2.709%**	39.30	6.605%
DPER-Pro	**0.9612**	2.793%	**40.94**	7.279%

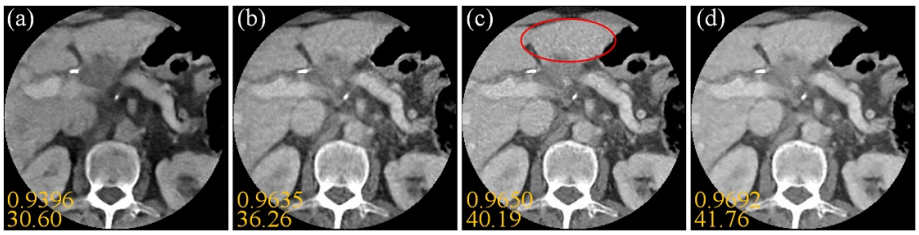


Figure 8. ROI reconstructions from simulated experiments (Poisson noise with 5×10^4 incident photons): FBPConvNet (a), DD_ExDenoise (b), DPER (c), and DPER-Pro (d). The display window is [0.016, 0.024]. SSIM and PSNR are in the bottom-left corner of each image.

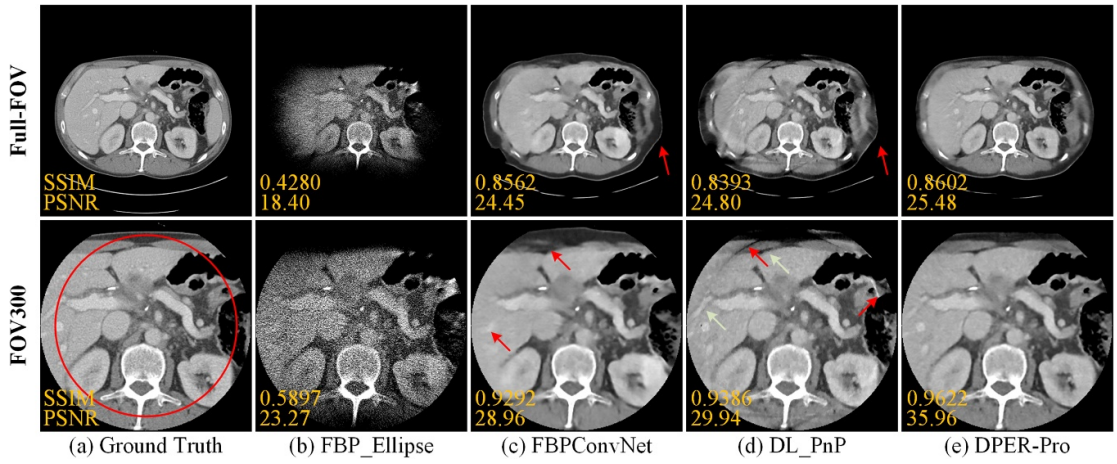


Figure 9. Extension of the recoverable region in simulated experiments: ground truth (a), FBP_Ellipse (b), FBPConvNet (c), DL_PnP (d), and DPER-Pro (e). The display window is [0.016, 0.024]. SSIM and PSNR are in the bottom-left corner of each image. The ROI is highlighted by the red circle in (a).

et al 2022). To evaluate extended reconstructions for exterior regions, we compared reconstructions of the full object and the FOV300 regions (see figure 9). Here, elliptical extrapolation was applied to the truncated projection before FBP reconstruction (denoted as 'FBP_Ellipse') which partially restored intensity and some anatomical structures in the exterior region (see figure 9(b)). FBPConvNet recovered some structures but missed many important details (see figure 9(c)). DL_PnP effectively restored detailed information within the FOV300 region (as indicated by the green arrows in figure 9(d)), but introduced ring-like artifacts at the ROI boundaries (see the red arrows in figure 9(d)). DPER-Pro achieved the best performance in the FOV300 region (see figure 9(e)). At the boundary region with severely missing projection, DL_PnP failed to correct the structural errors from FBPConvNet (see the full-FOV reconstructions in figures 9(c) and (d)). While DPER-Pro reconstructions had some residual errors, they outperformed other methods in restoring fine object details. The average quantitative metrics for the whole test set (summarized in table 4) further demonstrated the effectiveness of DPER-Pro in restoring structural information in the exterior region.

3.3. Real experiments

To assess the performance of the methods on real raw data, we conducted a scan on a human torso phantom using our laboratory cone-beam CT platform. In the scans, the x-ray source and detector were positioned 785 mm and 400 mm from the isocenter, respectively. The x-ray source operated at a tube voltage of 120 kV

Table 4. Quantitative metrics for extension of recoverable regions.

	SSIM		PSNR	
Metric	ROI300	Full-FOV	ROI300	Full-FOV
FBP_Ellipse	0.6371	0.4751	25.33	20.17
FBPConvNet	0.9297	0.8505	34.10	25.41
DL_PnP	0.9438	0.8305	35.71	26.03
DPER-Pro	**0.9603**	**0.8597**	**38.58**	**26.49**

with a 2 mm-thick aluminum filter. The size of detector elements was 0.45 mm after '3 × 3 binning'. Projections of 600 views uniformly distributed over 2π with two dose levels, low-dose and ultra-low dose, were collected for evaluation with the tube current being 7.1 mA and 1.8 mA, respectively. Reference images were generated by iterative reconstruction of projections acquired in an offset-detector mode where the pre-processing of projection data included an extra '2 × 3 averaging' operation to reduce noise. However, for interior tomography scans, the projection data did not undergo the aforementioned noise-reduction pre-processing. The reconstructed grid was 512 × 512, with a pixel size of 0.8 mm. To adapt to noisy data and geometric parameters, all DL models were trained under simulated datasets generated using the experimental configuration and simulated Poisson noise with 5×10^5 incident photons in projections.

3.3.1. ROI reconstruction

One exemplar slice is shown in figure 10(A) to illustrate ROI reconstructions. Beam-hardening artifacts are evident in the reference image. FBP reconstructions are noisy and with obvious intensity deviation. In the low-dose scan, DD_ExDenoise, DPER, and DPER-Pro achieved high-quality reconstructions. However, limitations in intensity restoration in specific regions (see the red annotations) for DD_ExDenoise can be observed. In the ultra-low-dose scan, DD_ExDenoise failed to restore the intensity and resulted in higher noise (as highlighted by the red elliptical region). While DPER demonstrated reasonable noise suppression capability, its performance declined compared to the low-dose scan. In contrast, DPER-Pro consistently delivered high-quality reconstructions with superior robustness across different dose levels.

3.3.2. Extension of recoverable regions

Extended reconstructions for the FOV300 region are shown in figure 10(B). Elliptical extrapolation significantly reduced intensity deviation in FBP_Ellipse reconstruction; however, the images remained noisy. FBPConvNet overestimated certain lung structures (as indicated by the red ellipse) which were partially corrected by DL_PnP. Nonetheless, DL_PnP's performance in exterior regions was limited. DPER-Pro effectively restored structural information in exterior regions, demonstrating its superior reconstruction capabilities.

3.4. Ablation study

We conducted ablation studies to evaluate the effects of dilated convolutions and progressive extension mechanism.

3.4.1. Effect of dilated convolutions

The projection-domain network (PDNet) in DPER and DPER-Pro utilizes dilated convolutions to expand the receptive field so that enables efficient identification of truncation features in the background projections. To validate this design, we compared the reconstructions of DPER with and without dilated convolutions. In this experiment, we modified only the structure of PDNet while keeping IDNet the same. Both dilated and non-dilated PDNets performed similarly on the training set. However, for test samples (see figure 11), DPER without dilated convolutions exhibited noticeable ring-shaped artifacts (as indicated by the red arrow in figure 11(d)). Direct reconstructions from the output of PDNet (i.e. the input of IDNet) further confirmed that these artifacts originated from the poor performance of PDNet and could not be fully eliminated by IDNet (see figure 11(c)). In contrast, DPER with dilated convolutions produced high-quality reconstructions (see figure 11(b)), which demonstrates that dilated convolutions effectively improve the reconstruction performance by enhancing the receptive field of PDNet.

3.4.2. Effect of the progressive extension mechanism

DPER-Pro employs a 'coarse-to-fine' strategy to progressively extend the recoverable region. To evaluate its effectiveness, we compared the results of a single-stage extension method (referred to as 'single stage extension') with those obtained from the first and second stages of DPER-Pro. The same slice from figure 9 is presented in figure 12. The average quantitative metrics for the ROI and FOV300 region across all test

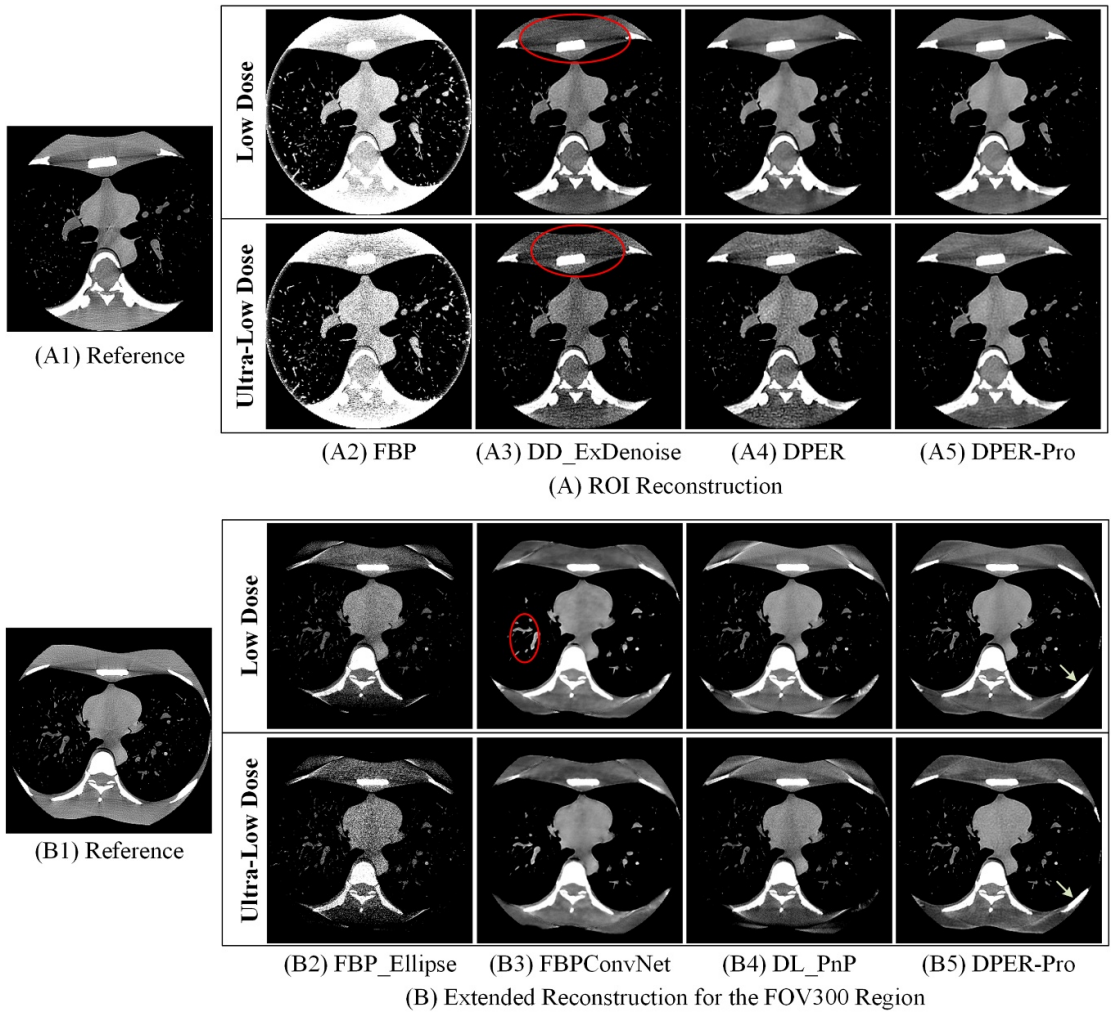


Figure 10. Results of practical CT scans. (A) ROI reconstruction; (B) extended reconstruction for FOV300 regions. The display window is [0.015, 0.024].

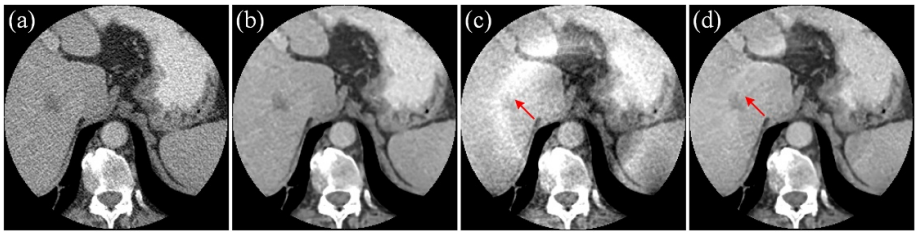


Figure 11. Ablation study on the dilated convolution: ground truth (a), DPER with dilated convolutions (b), direct reconstruction for the output of a non-dilated PDNet (c), and DPER without dilated convolutions (d). The display window is [0.016, 0.024].

samples are summarized in table 5. While the single-stage extension method achieved high-quality ROI reconstruction, it exhibited artifacts and structural distortions near the ROI boundaries (see the red arrows and circular region in figure 12(b)), along with significant structural loss outside the FOV300 region (see figure 12(a)). Reconstructions from the first stage of DPER-Pro (referred to as 'DPER-Pro (Stage1)') were slightly inferior to those of the single-stage extension method with ring-shaped structures near the ROI boundaries (as indicated by the red arrows in figure 12(c)). Nevertheless, the first stage partially restored the missing projection information in regions outside the ROI. Significant improvements were achieved in the second stage of DPER-Pro (referred to as 'DPER-Pro (Stage2)', see figure 12(d)), outperforming the single-stage extension method in both artifact suppression and structural restoration. Quantitative metrics further confirmed the effectiveness of the progressive extension mechanism.

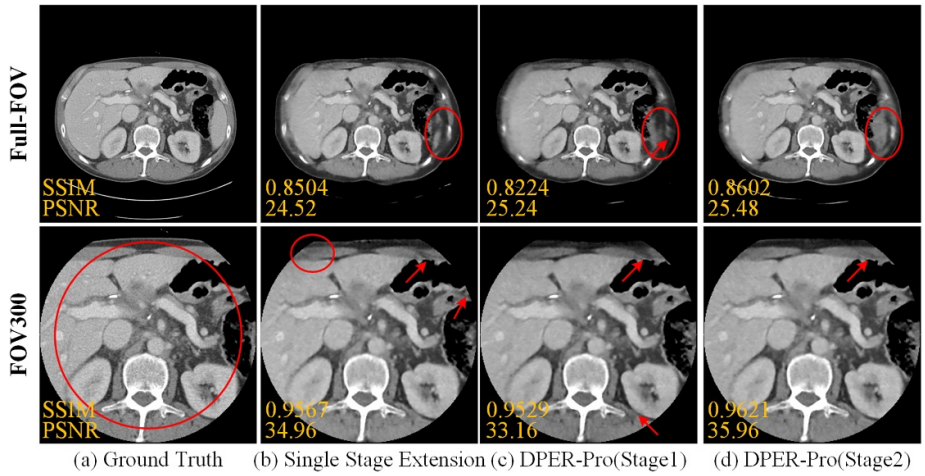


Figure 12. Ablation study on the progressive extension mechanism. The first row shows the full-FOV results, and the second row displays the FOV300 region results: ground truth (a), single-stage extension method (b), DPER-Pro (Stage1) (c), DPER-Pro (Stage2) (d). The display window is [0.016, 0.024]. SSIM and PSNR are in the bottom-left corner of each image.

Table 5. Quantitative metrics for ablation study of the progressive extension mechanism.

	SSIM		PSNR	
Metric	ROI	FOV300	ROI	FOV300
Single stage extension	0.9674	0.9574	41.29	37.79
DPER-Pro (Stage1)	0.9671	0.9564	40.34	37.09
DPER-Pro (Stage2)	**0.9681**	**0.9603**	**41.77**	**38.58**

4. Discussion

Low-dose interior tomography presents a coupled ill-posed problem involving noise contamination and data truncation which complicates image reconstruction. In this work, we conducted a comprehensive analysis of interior tomography by examining the composition of projection data and its angular sampling distribution. As illustrated in figure 2, the projection p_{IN} from a low-dose ROI scan consists of three main components: noise n_{p}, a non-truncated projection p_{ROI} from the ROI, and a truncated background projection $\overline{p}_{\mathrm{BG}}$ from regions outside the ROI. Since the FOV of scanners is typically centered on the ROI, the global angular sampling distribution is inherently non-uniform (see figure 5). While the ROI is sufficiently sampled, angular deficiency increases in exterior regions with the distance from the ROI center. The complex combination of projection components and non-uniform angular sampling presents big challenges for interior tomographic reconstruction. Aiming at high-quality ROI reconstruction and efficient extension of recoverable regions, we developed two novel methods DPER and DPER-Pro.

In general, ROI reconstruction is feasible if complete projection data covering p_{ROI} are obtained where p_{ROI} is the minimum required data. In the literature, extrapolation-based methods have been proposed to extend truncated projection outside the FOV, thereby acquiring a non-truncated projection that includes the full measurement of the exterior region. In contrast, we proposed a novel extraction-based reconstruction method DPER which estimates the contribution of the exterior region to interior projection measurement so that achieves p_{ROI} without sinogram extension. The key difference between DPER and extrapolation-based pathways lies in how the influence of unmeasured exterior regions is addressed. DPER primarily focuses on data inside the FOV which enables a joint noise suppression and extraction of truncated components. Our experimental results demonstrated the superiority of DPER in contextual structure recovery and noise tolerance compared to the extrapolation-based method DD_ExDenoise (see figures 7, 8 and 10). These findings underscore the strong capability of DPER in addressing the coupled ill-posed problem. An interesting finding is that PDNet largely utilizes the truncation property of data to distinguish $\overline{p}_{\mathrm{BG}}$ and p_{ROI}. Consequently, the expansion of the receptive field through dilated convolution is helpful in enhancing the projection extraction capability of PDNet (see figure 11).

Extension of recoverable regions reveals valuable information outside the ROI which is beneficial for localization and diagnosis. To address the challenge of non-uniform angular sampling and coupled components in interior projections, we further proposed a novel reconstruction method (DPER-Pro) that

employs a disentanglement process and a 'coarse-to-fine' strategy. Specifically, the various components in p_{IN} are firstly decoupled by three projection-domain subnetworks. The extracted background projection $\overline{p}_{BG}$, related to the exterior region, is then utilized for subsequent missing data compensation. Since the angular coverage of the exterior region varies drastically with its distance from the ROI center (see figure 5), DPER-Pro employs a two-stage manner guided by angular sampling distributions to progressively recover the missing data. Experiments demonstrated the effectiveness of DPER-Pro in extending the radius of the recoverable region by 17.2% to the FOV300 region without compromising ROI reconstruction quality (see figures 7, 9 and 10). The progressive extension mechanism also efficiently addressed the non-uniform data missing problem in exterior regions (see figure 12).

DPER and DPER-Pro address low-dose interior tomography from complementary perspectives. DPER employs a projection extraction strategy that combines denoising and truncation correction within a single network. DPER also efficiently disentangles the projection components of the ROI and exterior region which facilitates a fully disentangled learning process for DPER-Pro. While the noise tolerance of DPER is slightly inferior to DPER-Pro (see figures 8 and 10(A)), it significantly outperforms existing extrapolation-based methods. These findings underscore the advantages of combining efficient disentanglement and projection extraction within the DPER-Pro framework. Although DPER and DPER-Pro demonstrate strong potential in interior tomographic reconstruction, DPER-Pro's capability in the extension of recoverable regions is currently limited to the FOV300 region (with a minimum angular coverage of 120 degrees) in our experimental set-up. High-quality full-FOV reconstruction requires addressing more severe angular deficiencies, such as coverage below 60 degrees. Recent advancements in diffusion models have shown remarkable potential for limited-angle CT under extremely narrow angular coverage (Liu *et al* 2023, Zhang *et al* 2024). Future work could integrate diffusion models into the DPER-Pro framework to enable more efficient and robust extension, advancing high-quality full-FOV reconstructions.

5. Conclusion

In this work, we presented a comprehensive analysis of the composition of projection data and angular sampling distributions in low-dose interior tomography. Building on the concept of efficient disentanglement, we proposed two novel reconstruction pipelines: DPER, which extracts noise and background projections for accurate ROI reconstruction, and DPER-Pro, which employs a disentangled 'coarse-to-fine' strategy to achieve both high-quality ROI reconstruction and effective extension of recoverable regions. Experimental results on both simulated torso datasets and practical CT scans of a torso phantom demonstrated the superior performance of DPER and DPER-Pro in reconstructing reliable structures, improving generalization, and addressing the challenges of interior tomographic reconstruction. These methods provide a promising solution for advancing low-dose ROI imaging in medical applications.

Data availability statement

The data cannot be made publicly available upon publication because no suitable repository exists for hosting data in this field of study. The data that support the findings of this study are available upon reasonable request from the authors.

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